



Entering the Era of Discrete Diffusion Models: A Benchmark for Schrödinger Bridges and Entropic Optimal Transport

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Background: Entropic Optimal Transport and Schrödinger Bridge Problems

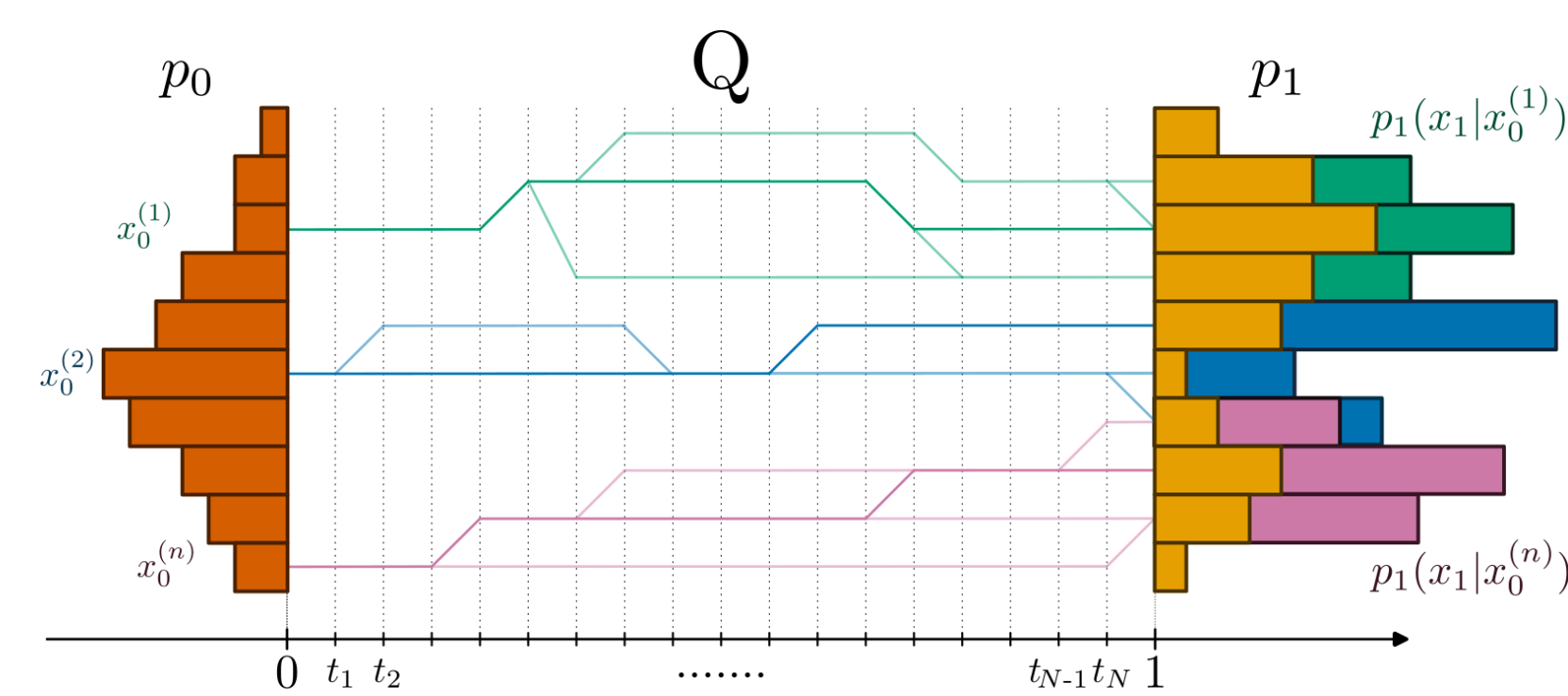
Problem setting. Given marginals p_0, p_1 on a discrete space $\mathcal{X}$ and a reference process q^{ref} , the dynamic and static formulations are equivalent views of the same entropic transport problem.

Dynamic Schrödinger Bridge (SB)

Seeks a stochastic process $q(x_0, x_{\text{in}}, x_1)$:

$$\min_{q \in \Pi_N(p_0, p_1)} \text{KL}(q(x_0, x_{\text{in}}, x_1) \parallel q^{\text{ref}}(x_0, x_{\text{in}}, x_1)).$$

$\Pi_N(p_0, p_1)$: processes with p_0 at $t=0$ and p_1 at $t=1$.

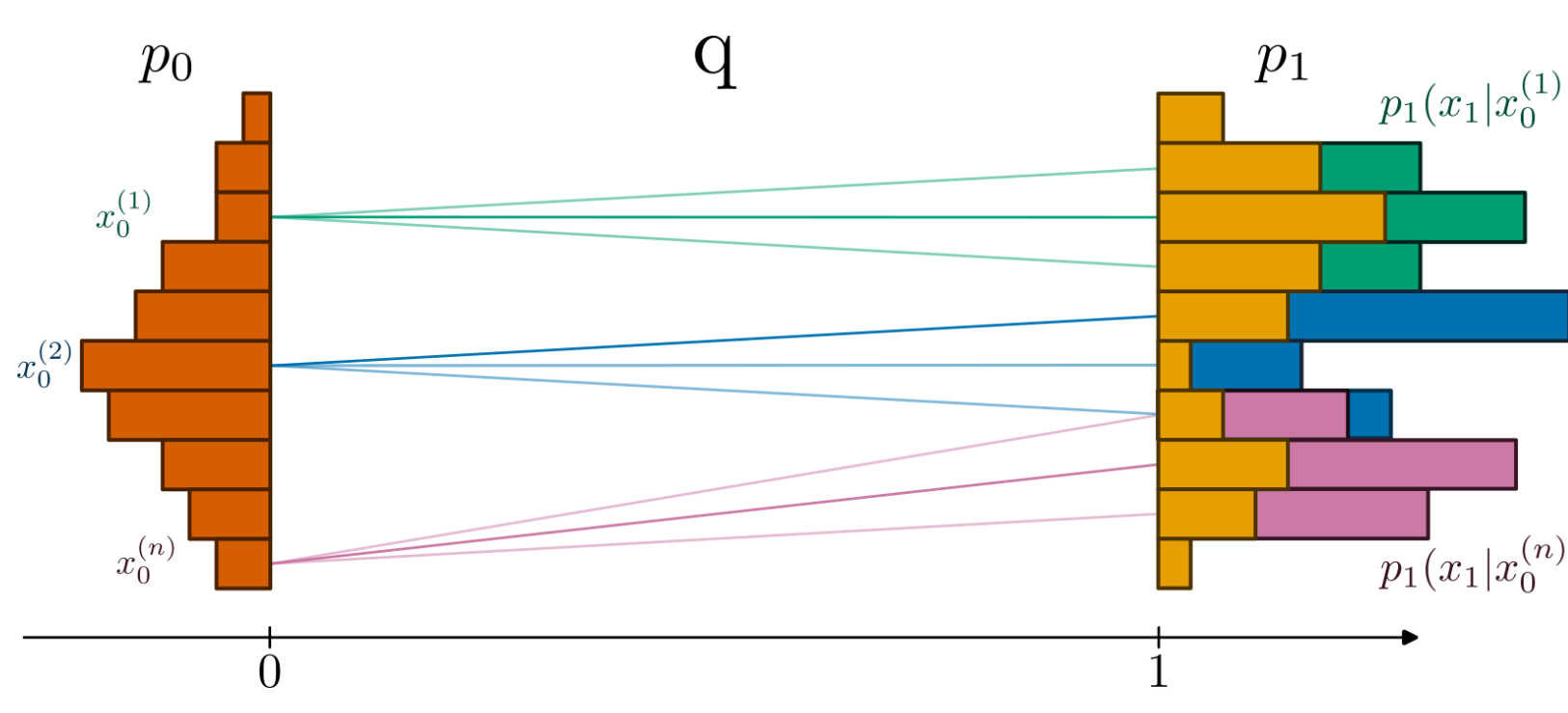


Static SB / Entropic Optimal Transport (EOT)

Seeks a joint distribution $q(x_0, x_1)$:

$$\begin{aligned} \min_{q \in \Pi(p_0, p_1)} \text{KL}(q(x_0, x_1) \parallel q^{\text{ref}}(x_0, x_1)), \\ \min_{q \in \Pi(p_0, p_1)} \mathbb{E}_q[-\log q^{\text{ref}}(x_1|x_0)] - H(q). \end{aligned}$$

$\Pi(p_0, p_1)$: joint distributions with marginals p_0, p_1 .



Motivation: Discrete and Continuous spaces

EOT/SB is well studied in continuous spaces ($\mathcal{X} = \mathbb{R}^D$), but remains underexplored in discrete spaces ($\mathcal{X} = \mathbb{S}^D$, $\mathbb{S} = \{0, \dots, S-1\}$): only CSBM is currently available, and no standard evaluation pipeline exists.

Goal. Establish the first benchmark for discrete-space EOT/SB solvers and expand the family of available solvers.

Benchmark: Reverse-Engineering Ground Truth

Key idea. Instead of fixing (p_0, p_1) and solving EOT/SB directly, we choose p_0 and construct $q^*(x_1 | x_0)$. This defines the joint $q^*(x_0, x_1) = p_0(x_0) q^*(x_1 | x_0)$, and hence induces p_1 through the following theorem.

Theorem (Benchmark Pair Construction for Discrete-Space EOT/SB). Let $p_0 \in \mathcal{P}(\mathcal{X})$ and $v^* : \mathcal{X} \rightarrow \mathbb{R}$. Consider a joint distribution $q^* \in \mathcal{P}(\mathcal{X}^2)$ such that $q^*(x_0) = p_0(x_0)$ and

$$q^*(x_1|x_0) \propto v^*(x_1) q^{\text{ref}}(x_1|x_0).$$

If $p_1(x_1) := q^*(x_1)$ is its second marginal, then q^* with q^{ref} defines the discrete-space EOT/SB between p_0 and p_1 .

Benchmark: Key Ingredients of Tractability

We parametrize v^* via the **Canonical Polyadic (CP) decomposition**:

$$v^*(x_1) = \sum_{k=1}^K \beta_k \prod_{d=1}^D r_k^d[x_1^d].$$

Universal approximation: On a finite space, CP approximates $v^*(x_1)$ arbitrarily well as K grows.

We use **factorizable** reference processes q^{ref} :

$$q^{\text{ref}}(x_1|x_0) = \prod_{d=1}^D q^{\text{ref}}(x_1^d|x_0^d).$$

- Gaussian q^{ref} :** favors nearby transitions; used for ordered spaces.
- Uniform q^{ref} :** assigns equal mass; used when no structure is assumed.

Benchmark: Resulting Parameterization

Conditional (one-step) distribution

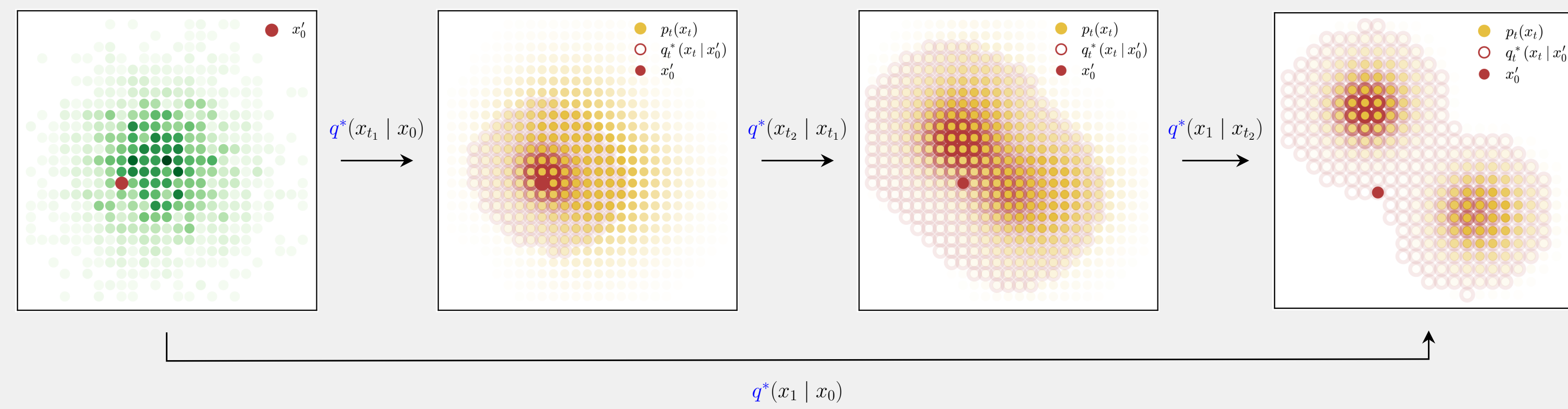
For static SB and EOT problems, we have:

$$\begin{aligned} q^*(x_1|x_0) &= \frac{1}{c^*(x_0)} \sum_{k=1}^K \beta_k \prod_{d=1}^D \left[r_k^d[x_1^d] q^{\text{ref}}(x_1^d|x_0^d) \right], \\ c^*(x_0) &= \sum_{k=1}^K \beta_k \prod_{d=1}^D \left(\sum_{x_1^d} r_k^d[x_1^d] q^{\text{ref}}(x_1^d|x_0^d) \right). \end{aligned}$$

Transition (multi-step) distributions

For the dynamic SB problem, we have:

$$\begin{aligned} q^*(x_{t_n}|x_{t_{n-1}}) &\propto q^{\text{ref}}(x_{t_n}|x_{t_{n-1}}) \sum_{k=1}^K \beta_k \prod_{d=1}^D u_{k,t_n}^d[x_{t_n}^d], \\ u_{k,t_n}^d[x_{t_n}^d] &= \sum_{x_1^d} q^{\text{ref}}(x_1^d|x_{t_n}^d) r_k^d[x_1^d]. \end{aligned}$$



Discrete-State EOT/SB solvers

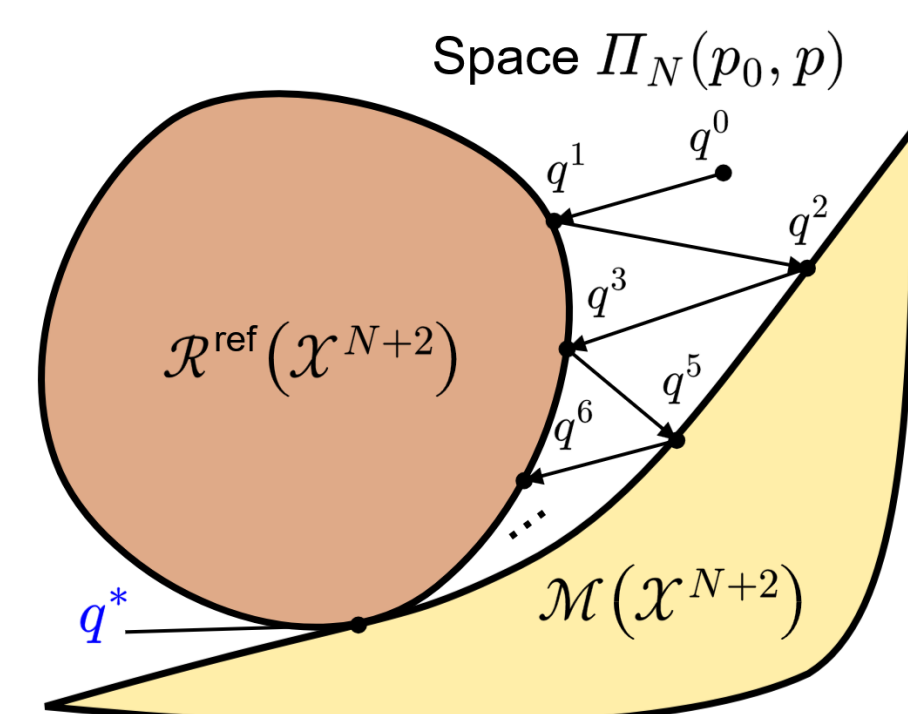
PRIOR: Categorical Schrödinger Bridge Matching (CSBM)

Idea. Alternating projections onto the Markov $\mathcal{M}$ and the reciprocal $\mathcal{R}^{\text{ref}}$ sets:

$$q^{2l+1} = \text{proj}_{\mathcal{M}}(q^{2l}), \quad q^{2l+2} = \text{proj}_{\mathcal{R}^{\text{ref}}}(q^{2l+1}).$$

Training. Bridge matching (BM) optimization:

$$\arg \min_{\theta} \mathbb{E}_{q^{2l+1}(x_0, x_1)} \left[\sum_{n=1}^N \mathbb{E}_{q_{n-1}^{\text{ref}}} \text{KL}(q_n^{\text{ref}} \parallel q_{\theta, n}) - \mathbb{E}_{q_N^{\text{ref}}} \log q_{\theta}(x_1 | x_{t_N}) \right].$$



NEW: α -Categorical Schrödinger Bridge Matching (α -CSBM)

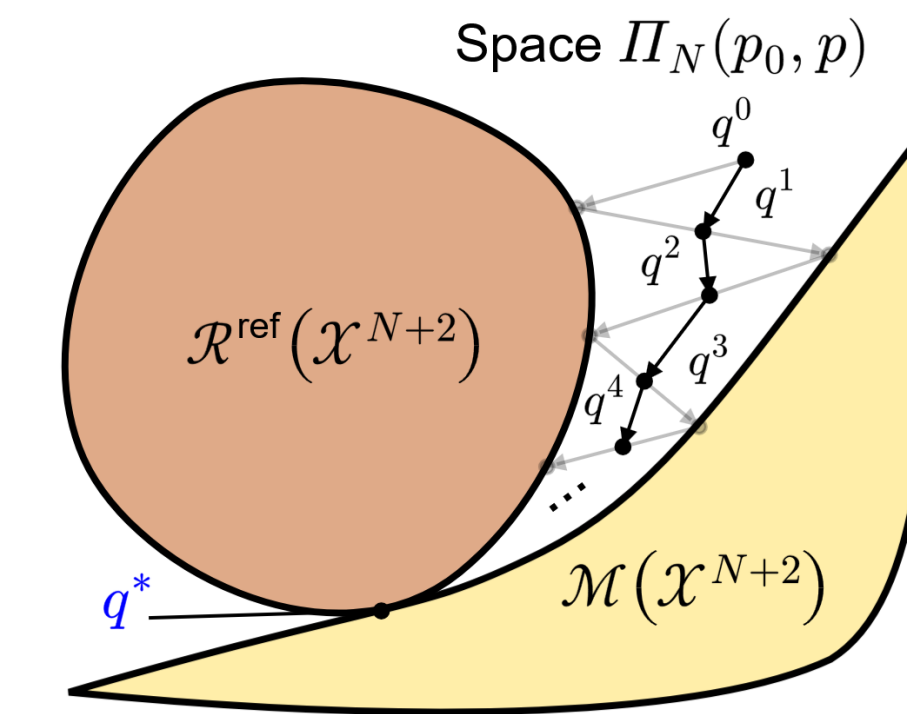
Idea. Relaxed projections with step size α :

$$q^{l+1} = (1 - \alpha)q^l + \alpha \text{proj}_{\mathcal{R}^{\text{ref}}}(\text{proj}_{\mathcal{M}}(q^l)).$$

Training. Joint forward/backward BM optimization:

$$\arg \min_{\theta} \frac{1}{2} \left(\text{KL}(\vec{r}_{\text{sg}} \parallel \vec{q}_{\theta}) + \text{KL}(\vec{r}_{\text{sg}}^{\leftarrow} \parallel \vec{q}_{\theta}^{\leftarrow}) \right),$$

where $r_{\text{sg}} = \text{proj}_{\mathcal{R}^{\text{ref}}}(q_{\theta})$.



NEW: Discrete Light Schrödinger Bridge (Matching) (DLightSB(-M))

Idea. DLightSB(-M) parameterizes v_{θ} so that q_{θ} always remains an SB. Hence, solving SB reduces to matching the target marginal p_1 via a **single projection**.

DLightSB

Focuses static target $q_{\theta}(x_1|x_0)$.

Training. Minimize $\text{KL}(q^* \parallel q_{\theta})$ via:

$$\arg \min_{\theta} \mathbb{E}_{p_0(x_0)} [\log c_{\theta}(x_0)] - \mathbb{E}_{p_1(x_1)} [\log v_{\theta}(x_1)].$$

DLightSB-M

Focuses dynamic target $q_{\theta}(x_{t_n}|x_{t_{n-1}})$.

Training. Minimize $\text{KL}(r \parallel q_{\theta})$ via BM optimization:

$$\arg \min_{\theta} \mathbb{E}_{q^{2l+1}(x_0, x_1)} \left[\sum_{n=1}^N \mathbb{E}_{q_{n-1}^{\text{ref}}} \text{KL}(q_n^{\text{ref}} \parallel q_{\theta, n}) - \mathbb{E}_{q_N^{\text{ref}}} \log q_{\theta}(x_1 | x_{t_N}) \right].$$

Evaluation Protocol

Baselines:

- Independent:** sample x_1 directly from p_1 ; ignores the joint distribution.
- Reference:** sample x_1 from q^{ref} ; follows prior transport.
- Feature-wise SB:** solve 1D SB independently for each dimension; ignores cross-dimensional dependence.

Metrics: Shape Score (SSM) and Trend Score (TSM) average over 1D and 2D marginals, respectively:

$$\text{SSM}^d = 1 - \frac{1}{2} \sum_{i=1}^{S-1} |\hat{q}^*(x^d) - \tilde{q}_{\theta}(x^d)|,$$

$$\text{TSM}^{d_m, d_n} = 1 - \frac{1}{2} \sum_{x^{d_m}=0}^{S-1} \sum_{x^{d_n}=0}^{S-1} |\hat{q}^*(x^{d_m}, x^{d_n}) - \tilde{q}_{\theta}(x^{d_m}, x^{d_n})|.$$

Here $\tilde{q}(x) = \frac{1}{|I|} \sum_{i \in I} \delta_{x^{(i)}}$ is the empirical distribution. Conditional variants average SSM and TSM over $x_0 \sim p_0$ and multiple x_1 per x_0 .

Conditional Trend Score Results

			$D = 2$				$D = 16$				$D = 64$			
			gaussian		uniform		gaussian		uniform		gaussian		uniform	
Method	Loss	$N+1$	0.02	0.05	0.005	0.01	0.02	0.05	0.005	0.01	0.02	0.05	0.005	0.01
<i>Independent</i>	-	-	0.471	0.779	0.521	0.551	0.569	0.676	0.369	0.458	0.478	0.509	0.351	0.432
<i>Reference</i>	-	-	0.085	0.277	0.230	0.273	0.111	0.082	0.090	0.095	0.195	0.121	0.136	0.110
<i>Feature-wise SB</i>	-	-	0.585	0.630	0.704	0.738	0.686	0.532	0.351	0.373	0.838	0.404	0.532	0.456
DLightSB	-	-	0.908	0.851	0.873	0.872	0.763	0.838	0.839	0.838	0.854	0.747	0.817	0.773
CSBM	KL	16	0.643	0.610	0.801	0.808	0.674	0.609	0.517	0.513	0.756	0.694	0.517	0.537
		64	0.837	0.794	0.840	0.846	0.801	0.738	0.572	0.624	0.788	0.698	0.486	0.536
	MSE	16	0.444	0.627	0.692	0.728	0.658	0.574	0.392	0.418	0.586	0.655	0.446	0.463
		64	0.227	0.748	0.686	0.673	0.726	0.716	0.482	0.551	0.496	0.714	0.479	0.478
α -CSBM	KL	16	0.588	0.616	0.819	0.814	0.610	0.659	0.545	0.557	0.773	0.659	0.484	0.515
		64	0.734	0.794	0.834	0.844	0.816	0.798	0.563	0.642	0.793	0.677	0.486	0.514
	MSE	16	0.470	0.607	0.767	0.781	0.604	0.649	0.449	0.512	0.592	0.618	0.448	0.457
		64	0.838	0.782	0.758	0.785	0.587	0.761	0.486	0.561	0.488	0.687	0.466	0.486
DLightSB-M	KL	16	0.817	0.848	0.860	0.864	0.702	0.820	0.831	0.820	0.731	0.622	0.468	0.500
		64	0.798	0.843	0.853	0.856	0.791	0.825	0.825	0.816	0.727	0.710	0.493	0.674
	MSE	16	0.624	0.839	0.755	0.838	0.451	0.824	0.783	0.785	0.532	0.683	0.645	0.328
		64	0.595	0.834	0.716	0.822	0.404	0.814	0.786	0.763	0.392	0.678	0.327	0.600

Key Takeaways

Baselines

- Reference** **degrades** because v^* strongly reweights the reference.
- Independent** **worsens** at low γ .
- Feature-wise SB** **degrades** as dimension increases.

These failures highlight the key challenges our benchmarks are designed to test.

DLightSB-like

- DLightSB: **strongest** performance across all setups.
- DLightSB-M: comparable results, **slight drop** due to KL.
- Both **benefit from the inductive bias** of the benchmark.

DLightSB(-M) can be viewed as **oracle-like** in this benchmark.

CSBM-like

- CSBM and α -CSBM perform **worse** than DLightSB(-M).
- α -CSBM matches CSBM quality at **lower computational cost**.
- Higher N generally improves metrics.
- KL consistently **outperforms** MSE (MSE over-smooths).